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1. Collect Data

We've chosen a stuff toy with distinct edges and features. See appendix A for more details.

2. Keypoint correspondences between images

Task part 1 a) Manual keypoint detection

Two OpenCV windows are opened and the keypoints of each images is stored upon clicking the image locations.

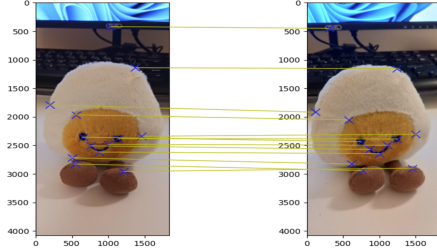


Figure 1. The keypoints are stored when clicking the cursor at a certain pixel area. Outlier percentage: 46.15 %.

Task part 1 b) Algorithmic keypoint detection

We compared the matching quality of 2 methods, MSE and SIFT.

Method 1: MSE

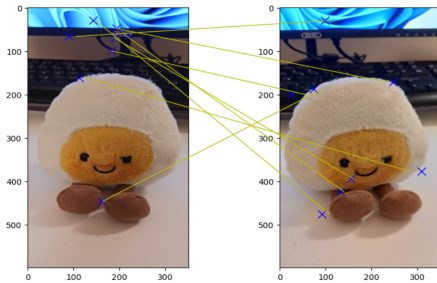


Figure 2. Points matched with MSE method. A square window is extracted from the source image and is used as a sliding convolution kernel applied to all the available points in the target image. The images are preprocessed to extract edges and corners using OpenCV Gaussian blur and canny edges. This is to locate keypoints and filter out unnecessary information. It is simple to implement, but is slow due to the convolution process, and inaccurate as it does not tolerate rotation or scaling. The outlier percentage is 50 %.

Method 2: SIFT

SIFT (Scale-invariant feature transform) [1] is a powerful algorithm designed for solving the correspondence problem. The key principle is to use multiple levels of Gaussian blur to extract the invariant points and calculate the horizontal and vertical gradients about the edges and corners. For details of the algorithm, please refer to the appendix.

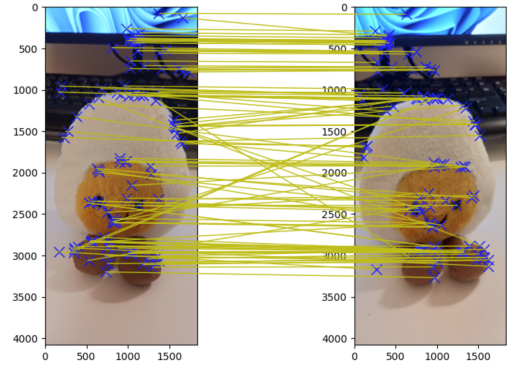


Figure 3. Points matched using SIFT algorithm. Outlier percentage: 44.72%

3. Camera calibration

Task part 1 Camera parameters extraction

$$K = \begin{pmatrix} 3454 & 0 & 1305.6 \\ 0 & 3291 & 1887 \\ 0 & 0 & 1 \end{pmatrix}$$

The camera extrinsic matrix is different for each chessboard image. We selected the first image's extrinsic parameter to display (see Appendix A for calibration details):

$$[R_0|t_0] = \begin{pmatrix} -0.02453 & -0.9662 & 0.2565 & -0.06513 \\ 0.9835 & 0.02268 & -0.1795 & -0.09985 \\ 0.1793 & -0.2567 & 0.94973135 & 0.5642 \end{pmatrix}$$

Task part 2 Camera distortion estimation

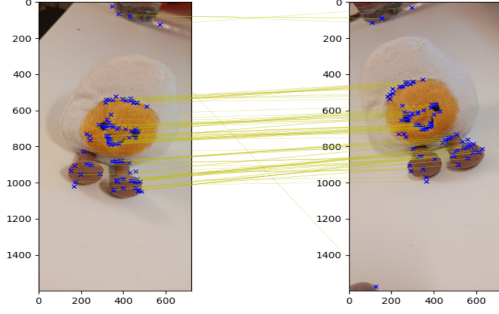
Radial distortion: $[k_1 = -0.040554, k_2 = 4.834, k_3 = -1.757]$

Tangential distortion: $[p_1 = -0.005689, p_2 = 0.10642]$

See Appendix B for visualization.

4. Transformation estimation

Task 1 a) Homography matrix



Homography keypoint correspondence matching. The RANSAC method is used over the 4 point method for higher accuracy. Almost all points are matched correctly.

$$H_{est.} = \begin{pmatrix} 0.6391 & 0.60213 & -360.088 \\ -0.6939 & 0.72039 & 316.9 \\ -1.948 \times 10^{-4} & -1.911 \times 10^{-5} & 1.000 \end{pmatrix}$$

Task 2 a and b) Fundamental matrix

Method 1: 8 point method

This method uses 8 corresponding points to form an 8x8 matrix. As the solutions were derived using few data points, it is more prone to errors and outliers.

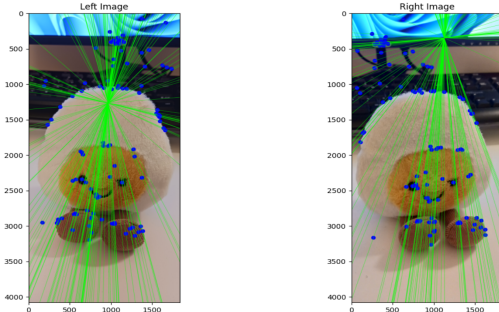


Figure 4. Epipolar line plot (8 point method)

Method 2: RANSAC

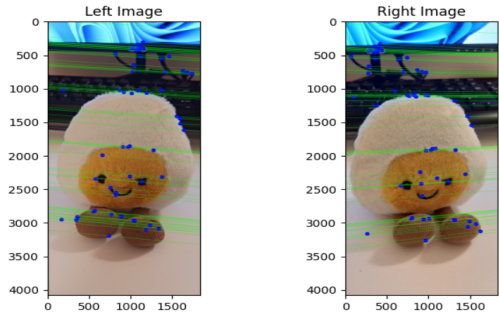


Figure 5. Epipolar line plot (RANSAC method). The method uses random sampling and model fitting making it robust to outliers. The epipolar lines now indicates the camera perspectives more accurately.

$$F_{est.} = \begin{pmatrix} -6.151 \times 10^{-7} & 1.2064 \times 10^{-5} & 0.009521 \\ 1.448 \times 10^{-5} & -5.0983 \times 10^{-7} & -0.1179 \\ -0.01513 & 0.10179 & 1.000 \end{pmatrix}$$

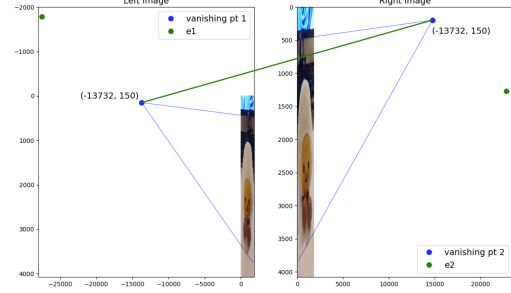


Figure 6. Horizon and vanishing point plot for the image pair

Task 3 Outlier Tolerance level

The RANSAC method theoretically tolerates (~50%) of outliers and produces more accurate estimate of the fundamental matrix. In practice, we've found that RANSAC produces good quality matching points even when the outlier exceeds 80 %.

5. 3D geometry

Task 1 Stereo rectification

Method 1: Uncalibrated

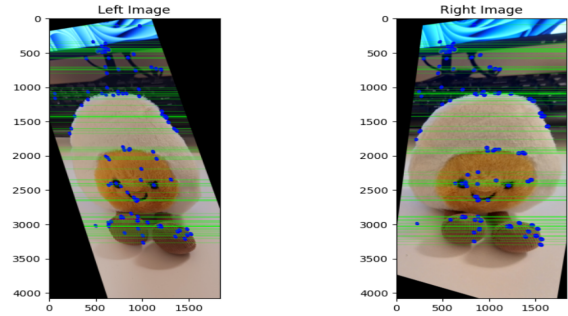


Figure 7. Stereo rectification pair (uncalibrated).

In this method, the camera extrinsics were not used and only the intrinsic parameters and the corresponding keypoints were used to estimate the homography projections. The advantage of this method is it relies on the robust estimation algorithms of OpenCV which produces more visually accurate result. The disadvantage is that we had no information of the relative transformation between the cameras; moreover, since it relies on the builtin functions, there's little control over the rectification process.

Method 2: Calibrated

The additional objects such bottle and computer screen was added for depth comparison. The calibration grid was used

to compute the 2 extrinsic matrices and extract the rotation and translation matrices, as well as to estimate the relative transformation between the cameras, and subsequently the 2 homography matrices for rectifying the 2 images. The advantage is that we had more control over the rectification parameters and the disadvantage is that there were more source of error arised from the estimation of the relative transform matrices.

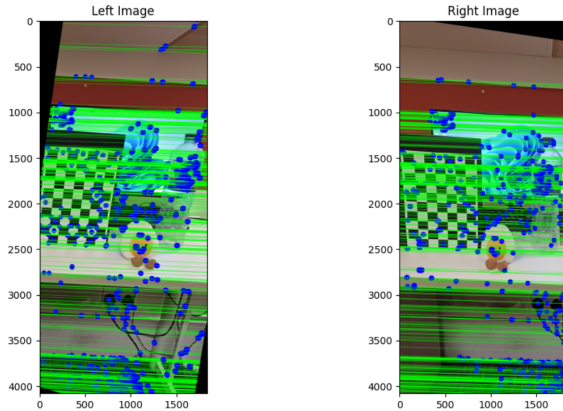
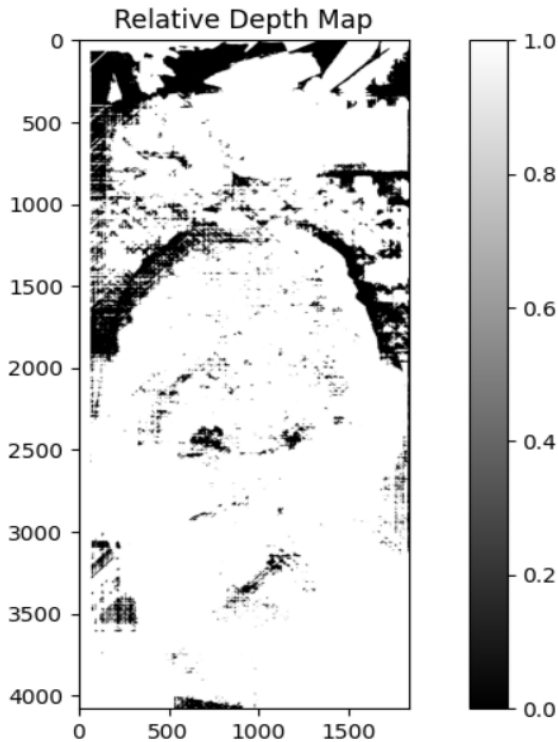


Figure 8. Stereo rectification pair computed using camera calibration and extrinsic matrices. As with previous steps, the matched keypoints, fundamental matrix and epipolar lines were computed for these images.

Task 2 Depth map



Depth map (uncalibrated rectification)

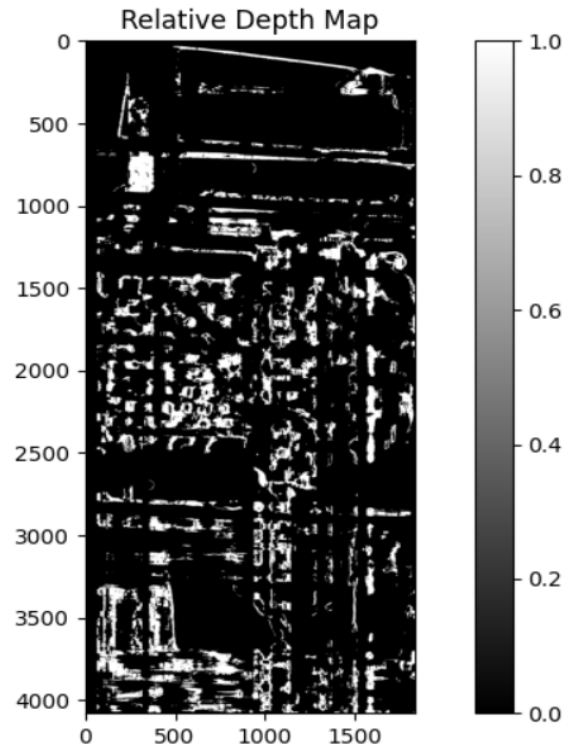


Figure 9. Depth map (calibrated rectification)

Both rectification methods produced good results for depth map estimation. Both methods also rectify the images so that the epipolar lines and the keypoints lie on the same horizontal line with minimal vertical disparity. In the depth computation, we used a window size of 100 and a block size of 45 reduce noise. We computed the relative depth where we assumed the baseline to be 1 because the exact spacing between the 2 camera centers were difficult to measure.

Conclusion

We have shown that SIFT combined with RANDAC method forms a robust stereo vision systems for both uncalibrated and calibrated images. And in which the uncalibrated method is able to perfectly align the epipolar lines. For the calibrated method, there are inevitable source of errors such as illumination changes, texture-less surfaces, or repetitive patterns. One could potentially reduce these errors by choosing better camera angles when performing camera calibration in order to obtain a more accurate intrinsic matrix; or one could use a more fundamental method to estimate the relative transformation matrix, for example, by extracting them from the essential matrix, or computing them from the 2 camera extrinsic matrices.

6. Appendix

Appendix A: data collection

The images are taken using a 12M pixel camera from a smart phone. All images are of size 1836x4080. The chessboard is printed from a printer with grid size 21mmx21mm. The photos are taken at different angles and locations to enable more accurate parameter estimation.



Figure 10. FD without calibration grid



Figure 11. FD with calibration grid

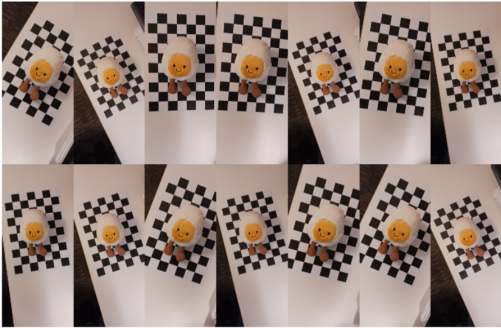


Figure 12. HG with calibration grid



Figure 13. HG without calibration grid



Figure 14. Multi-object scene for depth map estimation

In the camera calibration step, we used the OpenCV's built-in function to extract the chessboard corners from each images. For more accurate calibration results, we used 14 FD images that contain a 2D chessboard captured from various perspectives. The corresponding images coordinates and chessboard corners (the world coordinates) are then used to estimate the camera intrinsic matrix, K and the extrinsics matrix $R_i|t_i$.

We also calculated the camera matrices using the 6-point method. However, this method is prone to calibration errors such as errors between matching keypoints or world coordinate measurements. And the intrinsic matrix produced does not later on rectifies the images properly, hence we decided to use the OpenCV's built-in method.

Appendix B: Distortion estimation Following the camera calibration, we’ve extracted the distortion coefficients (See Task part 2). This is a comparison of the distorted (left) and undistorted (right) image computed using the distortion coefficients. Although by using the structural similarity index (SSIM), we computed the similarity between the 2 images as $SSIM = 0.9313$. Meaning the distortion coefficients have been successfully applied to undistort the image.



Figure 15. Sampled image after applying lens distortion parameters to undistort the image. Since the camera used has little lens distortion, the difference between the two images were small. Although by using the structural similarity index (SSIM), we computed the similarity between the 2 images as $SSIM = 0.9313$. Meaning the distortion coefficients have been successfully applied to undistort the image.

Appendix C: SIFT method overview [1]

The SIFT algorithm is outline in the following steps:

1. Extract the invariant key points using the difference of the Gaussian technique, where a Gaussian kernel of varying intensity is applied to he same image:

$$g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

2. Characterize each keypoints by computing the intensity gradients M_{ij} and rotation gradients R_{ij} :

$$M_{ij} = \sqrt{(x_{ij} - x_{i+1,j})^2 + (x_{ij} - x_{i,j+1})^2}$$

$$R_{ij} = atan2(x_{ij} - x_{i+1,j}, x_{i,j+1} - x_{ij})$$

3. Compute the keypoints descriptors using the location, scale and rotation estimated of each keypoints.
4. Match the keypoint descriptors to compute matching keypoints.

Appendix D: Comparison of object types on rectification results

The rectification process is likely to be largely affected by the properties of the chosen object, such as it’s pattern color gradient and color intensity. Although the in main report, we’ve been focused on a stuffed toy, we’ve also discovered interesting results with other objects.

Beverage can:

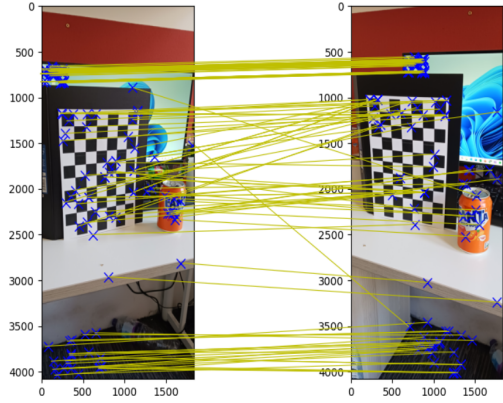


Figure 16. Keypoints correspondence matched with a can object.

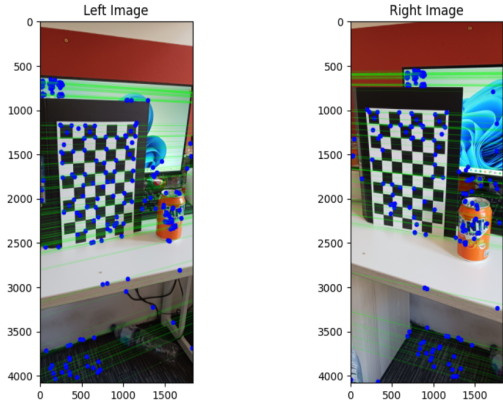


Figure 17. Epipolar lines calculated for the can object.

We observed that the quality of the matched keypoints, epipolar lines and the rectified images were equally good. The relative depth map produced was even slightly more distinctive than the one estimated using the stuffed toy. These results not only demonstrated that our computer vision pipeline is robust different variety of objects, but also the influence of the object’s property on the computation results of a computer vision system.

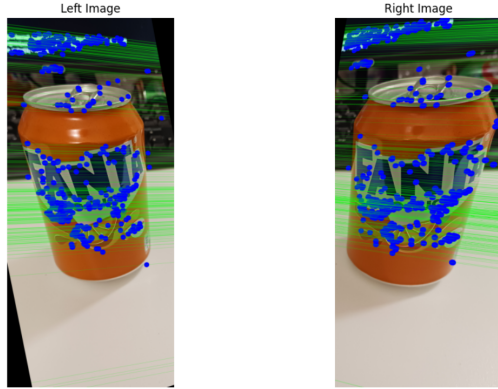


Figure 18. Stereo rectification results for the can object

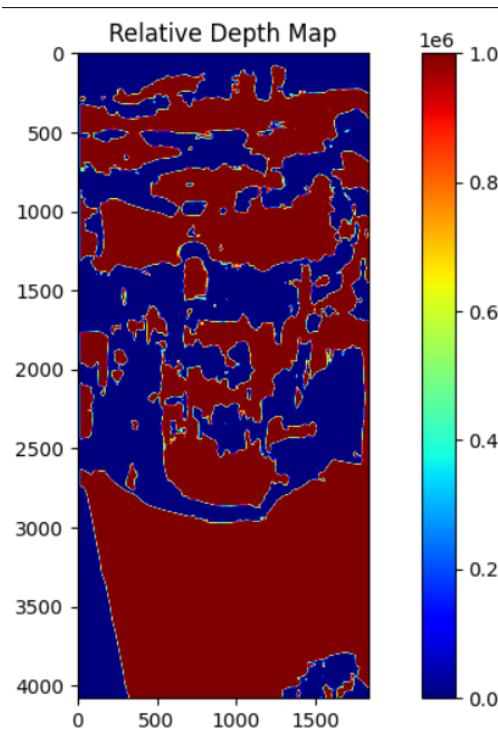


Figure 19. Depth map for the can object. Red areas (higher intensity) indicates closer distance and blue areas (lower intensity) indicates further distance.

References

- [1] D. Lowe. Object recognition from local scale-invariant features. In *Proceedings of the Seventh IEEE International Conference on Computer Vision*, volume 2, pages 1150–1157 vol.2, 1999.